

Homework 1

TCP Client/Server Programming & Packet Analysis

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Outline

- Client & Server source code
- Wireshark screenshots with labels and description
 - TCP three-way handshake packets
 - First Application data packet
 - RTT (Round Trip Time)
 - Connection Close Packets
 - Time Sequence Diagram

Client & Server source code

Server source code

```
1  from socket import *
2  serverPort = 12000
3  serverSocket = socket(AF_INET,SOCK_STREAM) # Create a TCP wellcoming socket
4  serverSocket.bind(('',serverPort)) # Assign IP address and port number to socket
5  serverSocket.listen(1) # Listen for TCP connections
6  print('The server is ready to receive')
7  while True:
8      connctionSocket, addr = serverSocket.accept() # Accept TCP connection from the client
9      while True:
10         sentence = connctionSocket.recv(1024).decode() # Receive the sentence from the client
11         if not sentence: # Client closed connection
12             break
13         if sentence == "exit":
14             connctionSocket.close() # Close the connection and send FIN to the client
15             break
16         num = sentence.split(",")
17         result = int(num[0]) * int(num[1]) # Calculate the multiple of the two numbers
18         connctionSocket.send(str(result).encode()) # Send the result back to the client
```

Client source code

```
1  from socket import *
2  serverName = '127.0.0.1' # server name is local host
3  serverPort = 12000
4  clientSocket = socket(AF_INET, SOCK_STREAM) # Create a TCP socket
5  clientSocket.connect((serverName, serverPort)) # Connect to the server
6  while True:
7      sentence = input('Input two numbers separated by a comma (or exit): ') # Get user input
8      clientSocket.send(sentence.encode()) # Send the sentence to the server
9      if sentence == "exit":
10         # Wait for the server to close the connection and send FIN
11         data = clientSocket.recv(1024)
12         if not data:
13             print('Connection closed by the server.')
14             clientSocket.close()
15             break
16         modifiedSentence = clientSocket.recv(1024).decode() # Receive the modified sentence from the server
17         print('The multiple of two number is: ', modifiedSentence) # Print the multiple of the two numbers
```

Wireshark screenshots with labels and description

Wireshark Screenshot

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	56	58468 → 12000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM=1
2	0.000098600	127.0.0.1	127.0.0.1	TCP	56	12000 → 58468 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM=1
3	0.000127200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=1 Ack=1 Win=65280 Len=0
4	4.307345100	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.307416700	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.307586600	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.307632400	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0
8	7.814351900	127.0.0.1	127.0.0.1	TCP	49	58468 → 12000 [PSH, ACK] Seq=4 Ack=3 Win=65280 Len=5
9	7.814387800	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=3 Ack=9 Win=65280 Len=0
10	7.814478400	127.0.0.1	127.0.0.1	TCP	48	12000 → 58468 [PSH, ACK] Seq=3 Ack=9 Win=65280 Len=4
11	7.814516500	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=9 Ack=7 Win=65280 Len=0
12	12.804186600	127.0.0.1	127.0.0.1	TCP	49	58468 → 12000 [PSH, ACK] Seq=9 Ack=7 Win=65280 Len=5
13	12.804218400	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=7 Ack=14 Win=65280 Len=0
14	12.804433300	127.0.0.1	127.0.0.1	TCP	48	12000 → 58468 [PSH, ACK] Seq=7 Ack=14 Win=65280 Len=4
15	12.804470400	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=14 Ack=11 Win=65280 Len=0
16	16.481269400	127.0.0.1	127.0.0.1	TCP	48	58468 → 12000 [PSH, ACK] Seq=14 Ack=11 Win=65280 Len=4
17	16.481341200	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=11 Ack=18 Win=65280 Len=0
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0
20	16.481711100	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [FIN, ACK] Seq=18 Ack=12 Win=65280 Len=0
21	16.481767000	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=12 Ack=19 Win=65280 Len=0

Description

- TCP three-way handshake packets
- First Application data packet
- RTT (Round Trip Time)
- Connection Close Packets
- Time Sequence Diagram

TCP three-way handshake packets

Packet No.1: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
1	0.00000...	127.0.0.1	127.0.0.1	TCP	56	58468 → 12000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
2	0.00009...	127.0.0.1	127.0.0.1	TCP	56	12000 → 58468 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256
3	0.00012...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=1 Ack=1 Win=65280 Len=0
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.30763...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0
8	7.81435...	127.0.0.1	127.0.0.1	TCP	49	58468 → 12000 [PSH, ACK] Seq=4 Ack=3 Win=65280 Len=5
9	7.81438...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=3 Ack=9 Win=65280 Len=0

Frame 1: 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface 0

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

- 0100 = Version: 4 → 使用 IPv4 進行定址與路由
- 0101 = Header Length: 20
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 52
- Identification: 0xb5a4 (46500)
- 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6) → 使用 TCP 進行可靠傳輸
- Header Checksum: 0x471d [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 127.0.0.1 → Client 端的 IP
- Destination Address: 127.0.0.1 → Server 端的 IP
- [Stream index: 0]

Transmission Control Protocol, Src Port: 58468, Dst Port: 12000, Seq: 0, Win: 65535, Len: 0

Internet Protocol Version 4 (ip), 20 byte(s)

Packets: 21

Profile: Default

TCP three-way handshake packets

Packet No.1: Client → Server

The image shows a Wireshark packet capture of the first packet in a TCP three-way handshake. The packet list at the top shows seven packets, with the first packet being the SYN packet from 127.0.0.1 to 127.0.0.1 on port 58468 to 12000. The packet details pane on the left shows the Transmission Control Protocol fields, and the packet bytes pane on the right shows the raw data. Red arrows and text boxes highlight key fields and their meanings.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.00000...	127.0.0.1	127.0.0.1	TCP	56	58468 → 12000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
2	0.00009...	127.0.0.1	127.0.0.1	TCP	56	12000 → 58468 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256
3	0.00012...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=1 Ack=1 Win=65280 Len=0
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.30763...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0

Transmission Control Protocol

- Source Port: 58468 → Client 端隨機分配的 Port
- Destination Port: 12000 → Server 端監聽的 Port
- [Stream index: 0]
- [Stream Packet Number: 1]
- [Conversation completeness: 0]
- [TCP Segment Len: 0] → 傳遞資料長度為 0
- Sequence Number: 0 (relative sequence number) → Client 端的 Seq #，初始值設定為 0，表示目前要傳遞給 Server 端的 Byte Stream 計數
- Sequence Number (raw): 3074361385
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 0 → Client 端的 Ack # 還不存在
- Acknowledgment number (raw): 0
- 1000 = Header Length: 32 bytes (8)
- Flags: 0x002 (SYN) → Flag 中僅有 SYN 的位元被標記，表示其為建立連線（三向交握）的第一個封包，也就是 Client 向 Server 傳送請求建立連線的訊息
- Window: 65535
- [Calculated window size: 65535]
- Checksum: 0xa43c [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- Options: (12 bytes), Maximum segment size, No-Operation (NOP),
- [Timestamps]

Flags: 0x002 (SYN)

- 000. = Reserved: Not set
- ...0. = Accurate ECN: Not set
- ...0... = Congestion Window Reduced: Not set
- ...0... = ECN-Echo: Not set
- ...0... = Urgent: Not set
- ...0... = Acknowledgment: Not set
- ...0... = Push: Not set
- ...0... = Reset: Not set
- ...1... = Syn: Set
- ...0... = Fin: Not set
- [TCP Flags:S.]

Transmission Control Protocol (tcp), 32 byte(s) Packets: 21 Profile: Default

TCP three-way handshake packets

Packet No.2: Server → Client

No.	Time	Source	Destination	Protocol	Length	Info
1	0.00000...	127.0.0.1	127.0.0.1	TCP	56	58468 → 12000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
2	0.00009...	127.0.0.1	127.0.0.1	TCP	56	12000 → 58468 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256
3	0.00012...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=1 Ack=1 Win=65280 Len=0
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.30763...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0

Frame 2: 56 bytes on wire (448 bits), 56 bytes captured (448 bits)

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

0100 = Version: 4

.... 0101 = Header Length: 20

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 52

Identification: 0xb5a5 (46501)

010. = Flags: 0x2, Don't fragment

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 128

Protocol: TCP (6)

Header Checksum: 0x471c [validation disabled]

[Header checksum status: Unverified]

Source Address: 127.0.0.1

Destination Address: 127.0.0.1

[Stream index: 0]

Transmission Control Protocol, Src Port: 12000, Dst Port: 58468, S

0000 02 00 00 00 45 00 00 34 b5 a5 40 00 80 06 47 1c

0010 7f 00 00 01 7f 00 00 01 2e e0 e4 64 fd fd 57 f1

0020 b7 3f 08 2a 80 12 ff ff 4e 3c 00 00 02 04 ff d7

0030 01 03 03 08 01 01 04 02

E..4 ..@...G.

..... .d.W.

?.*..... N<.....

.....

使用 IPv4 進行定址與路由

使用 TCP 進行可靠傳輸

Server 端的 IP

Client 端的 IP

TCP three-way handshake packets

Packet No.2: Server → Client

The image shows a Wireshark packet capture of the second packet in a TCP three-way handshake. The packet is a SYN-ACK from the server to the client. The packet details pane shows the following fields:

- Source Port:** 12000 (Server端監聽的 Port)
- Destination Port:** 58468 (Client端隨機分配的 Port)
- Sequence Number:** 0 (relative sequence number) (Server端的 Seq #, 表示目前傳遞給 Client 端的 Byte Stream 的起始序號)
- Acknowledgment Number:** 1 (relative ack number) (Ack = 1 表示 Server 已成功接收 Client 的 SYN 請求 (Seq 0), 並期待 Client 下一次發送從序號 1 開始的資料。)
- Flags:** 0x012 (SYN, ACK) (Flag 中 SYN 與 ACK 的位元被標記, 其中的 ACK 位元表示 Server 已收到來自 Client 想要建立連結的訊息, SYN 位元用於同步 Server 端的初始序號)

The packet bytes pane shows the raw data of the packet, including the header and payload. The header fields are highlighted in blue, and the payload is shown in hexadecimal and ASCII.

TCP three-way handshake packets

Packet No.3: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
1	0.00000...	127.0.0.1	127.0.0.1	TCP	56	58468 → 12000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
2	0.00009...	127.0.0.1	127.0.0.1	TCP	56	12000 → 58468 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256
3	0.00012...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=1 Ack=1 Win=65280 Len=0
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2

Frame 3: 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface 0

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

- 0100 = Version: 4 → 使用 IPv4 進行定址與路由
- 0101 = Header Length: 20
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 40
- Identification: 0xb5a6 (46502)
- 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6) → 使用 TCP 進行可靠傳輸
- Header Checksum: 0x4727 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 127.0.0.1 → Client 端的 IP
- Destination Address: 127.0.0.1 → Server 端的 IP
- [Stream index: 0]

Transmission Control Protocol, Src Port: 58468, Dst Port: 12000, Seq: 1, Win: 65280, Len: 0

0000 02 00 00 00 45 00 00 28 b5 a6 40 00 80 06 47 27E..(..@...G'
0010 7f 00 00 01 7f 00 00 01 e4 64 2e e0 b7 3f 08 2ad...?.*
0020 fd fd 57 f2 50 10 00 ff 88 34 00 00 ...W.P... .4..

TCP three-way handshake packets

Packet No.3: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
1	0.00000...	127.0.0.1	127.0.0.1	TCP	56	58468 → 12000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
2	0.00009...	127.0.0.1	127.0.0.1	TCP	56	12000 → 58468 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256
3	0.00012...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=1 Ack=1 Win=65280 Len=0
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2

Transmission Control Protocol, Src Port: 58468, Dest Port: 12000, Seq: 1, Len: 0

Source Port: 58468 → Client 端隨機分配的 Port

Destination Port: 12000 → Server 端監聽的 Port

[Stream index: 0]

[Stream Packet Number: 3]

[Conversation completeness: 1]

[TCP Segment Len: 0] → 傳遞資料長度為 0 (Len=0)

Sequence Number: 1 (relative sequence number) → Client 端的 Seq # 為 1，因為前面的 SYN 封包已經消耗一個序號，而下一個資料傳遞的序號編號為 1

Sequence Number (raw): 3074361386

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number) → Ack = 1 表示 Client 已成功收到 Server 回傳的 SYN+ACK (Seq 0)，並期待 Server 下一次發送從序號 1 開始的資料

Acknowledgment number (raw): 4261238770

0101 = Header Length: 20 bytes (5)

Flags: 0x010 (ACK) → Flag 中 ACK 的位元被標記，表示 Client 已收到來自 Server 端回覆的訊息，也就是正式建立好連線，後續可以直接進行資料傳遞

Window: 255

[Calculated window size: 65280]

[Window size scaling factor: 256]

Checksum: 0x8834 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

[SEQ/ACK analysis]

0000 02 00 00 00 45 00 00 28 b5 a6 40 00 80 06 47 27 ...E..(..@...G'

0010 7f 00 00 01 7f 00 00 01 e4 64 2e e0 b7 3f 08 2ad...?.*

0020 fd fd 57 f2 50 10 00 ff 88 34 00 00 ...W.P...4..

Flags: 0x010 (ACK)

000. = Reserved: Not set

...0 = Accurate ECN: Not set

.... 0... = Congestion Window Reduced: Not set

.... .0.. = ECN-Echo: Not set

.... ..0. = Urgent: Not set

.... ...1 = Acknowledgment: Set

....0... = Push: Not set

....0.. = Reset: Not set

....0.. = Syn: Not set

....0.. = Fin: Not set

[TCP Flags:A....]

First Application data packet

Client 端輸入：5,7

Server 端回傳：35

```
Microsoft Windows [版本 10.0.26200.8037]
(c) Microsoft Corporation. 著作權所有，並保留一切權利。

C:\Users\suyou>cd ./Desktop/Computer Network HW/Assignment 1

C:\Users\suyou\Desktop\Computer Network HW\Assignment 1>python .\TCPClient.py
Input two numbers separated by a comma (or exit): 5,7
The multiple of two number is: 35
Input two numbers separated by a comma (or exit): 11,98
The multiple of two number is: 1078
Input two numbers separated by a comma (or exit): 55,72
The multiple of two number is: 3960
Input two numbers separated by a comma (or exit): exit
Connection closed by the server.

C:\Users\suyou\Desktop\Computer Network HW\Assignment 1>
```

First Application data packet

Packet No.4: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.30763...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0

▶ Frame 4: 47 bytes on wire (376 bits), 47 bytes captured (376 bits) on Null/Loopback	0000	02 00 00 00 45 00 00 2b b5 f1 40 00 80 06 46 d9	E..+..@...F
▶ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1	0010	7f 00 00 01 7f 00 00 01 e4 64 2e e0 b7 3f 08 2a	d...?..*
0100 = Version: 4	0020	fd fd 57 f2 50 18 00 ff 1b fd 00 00 35 2c 37	..W.P...5,7
.... 0101 = Header Length: 20 (bytes)			
▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)			
Total Length: 43			
Identification: 0xb5f1 (46577)			
▶ 010. = Flags: 0x2, Don't fragment			
...0 0000 0000 0000 = Fragment Offset: 0			
Time to Live: 128			
Protocol: TCP (6)			
Header Checksum: 0x46d9 [validation disabled]			
[Header checksum status: Unverified]			
Source Address: 127.0.0.1			
Destination Address: 127.0.0.1			
[Stream index: 0]			
▶ Transmission Control Protocol, Src Port: 58468, Dst Port: 12000, Seq: 1, Win: 65280, Len: 3			
▶ Data (3 bytes)			

First Application data packet

Packet No.4: Client → Server

The image shows a Wireshark packet capture of a TCP segment. The packet list at the top shows four packets. Packet 4 is selected, showing its details and raw data. Red arrows point from specific fields to explanatory text boxes.

No.	Time	Source	Destination	Protocol	Length	Info
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.30763...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0

Transmission Control Protocol, Src Port: 58468, Dst Port: 12000

- Source Port: 58468 → Client 端隨機分配的 Port
- Destination Port: 12000 → Server 端監聽的 Port
- [Stream index: 0]
- [Stream Packet Number: 4]
- [Conversation completeness: Complete]
- [TCP Segment Len: 3] → 傳遞資料長度為 3
- Sequence Number: 1 (relative sequence number) → Client 端的 Seq #, 表示目前傳遞給 Server 端的 Byte Stream 的起始序號
- Sequence Number (raw): 3074361386
- [Next Sequence Number: 4 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number) → Ack = 1 表示 Client 已成功收到 Server 傳遞的 Seq 0 以前的所有訊息, 並期待 Server 下一次發送從序號 1 開始的資料
- Acknowledgment number (raw): 4261238770
- 0101 = Header Length: 20 bytes (5)
- Flags: 0x018 (PSH, ACK) → Flag 中 ACK、PSH 的位元被標記, ACK位元表示 Client 已收到先前 Server 端傳遞的資料, PSH 位元則是表示封包攜帶了應用層的資訊, 需要立即傳遞至應用程式進行資料處理
- Window: 255
- [Calculated window size: 65280]
- [Window size scaling factor: 256]
- Checksum: 0x1bfd [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- [Timestamps]
- [SEQ/ACK analysis]
- TCP payload (3 bytes)
- Data (3 bytes)

Flags: 0x018 (PSH, ACK)

- 000. = Reserved: Not set
- ...0. = Accurate ECN: Not set
- ...0. = Congestion Window Reduced: Not set
- ...0. = ECN-Echo: Not set
- ...0. = Urgent: Not set
- ...1. = Acknowledgment: Set
- ...1. = Push: Set
- ...0. = Reset: Not set
- ...0. = Syn: Not set
- ...0. = Fin: Not set
- [TCP Flags:AP...]

First Application data packet

Packet No.4: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
2	0.00009...	127.0.0.1	127.0.0.1	TCP	56	12000 → 58468 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=25
3	0.00012...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=1 Ack=1 Win=65280 Len=0
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0

Destination Port: 12000
[Stream index: 0]
[Stream Packet Number: 4]
▸ [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 3]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 3074361386
[Next Sequence Number: 4 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 4261238770
0101 = Header Length: 20 bytes (5)
▸ Flags: 0x018 (PSH, ACK)
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0x1bfd [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
▸ [Timestamps]
▸ [SEQ/ACK analysis]
TCP payload (3 bytes)
▾ Data (3 bytes)
Data: 352c37
[Length: 3]

0000 02 00 00 00 45 00 00 2b b5 f1 40 00 80 06 46 d9 ...E.+..@...F.
0010 7f 00 00 01 7f 00 00 01 e4 64 2e e0 b7 3f 08 2ad...?.*
0020 fd fd 57 f2 50 18 00 ff 1b fd 00 00 35 2c 37 ...W.P...5,7

Client 實際上傳遞給 Server 的資料

First Application data packet

Packet No.5: Server → Client

No.	Time	Source	Destination	Protocol	Length	Info
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.30763...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0
8	7.81435...	127.0.0.1	127.0.0.1	TCP	49	58468 → 12000 [PSH, ACK] Seq=4 Ack=3 Win=65280 Len=5

Frame 5: 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface 0

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

0100 = Version: 4

.... 0101 = Header Length: 20 bytes

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 40

Identification: 0xb5f2 (46578)

010. = Flags: 0x2, Don't fragment

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 128

Protocol: TCP (6)

Header Checksum: 0x46db [validation disabled]

[Header checksum status: Unverified]

Source Address: 127.0.0.1

Destination Address: 127.0.0.1

[Stream index: 0]

Transmission Control Protocol, Src Port: 12000, Dst Port: 58468, Seq: 12000, Win: 65280, Len: 0

Annotations:

- 使用 IPv4 進行定址與路由 (points to IP version 4)
- 使用 TCP 進行可靠傳輸 (points to Protocol: TCP)
- Server 端的 IP (points to Source Address: 127.0.0.1)
- Client 端的 IP (points to Destination Address: 127.0.0.1)

First Application data packet

Packet No.5: Server → Client

The image displays a Wireshark packet capture analysis of Packet No. 5: Server → Client. The packet list shows four packets, with Packet 5 being the first application data packet. The packet details pane shows the following fields:

- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- Transmission Control Protocol, Src Port: 12000, Destination Port: 58468
- [Stream index: 0]
- [Stream Packet Number: 5]
- [Conversation completeness: Complete]
- [TCP Segment Len: 0]
- Sequence Number: 1 (relative sequence number)
- Sequence Number (raw): 4261238770
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 4 (relative ack number)
- Acknowledgment number (raw): 3074361389
- 0101 = Header Length: 20 bytes (5)
- Flags: 0x010 (ACK)
- Window: 255
- [Calculated window size: 65280]
- [Window size scaling factor: 256]
- Checksum: 0x8831 [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- [Timestamps]
- [SEQ/ACK analysis]

The packet bytes pane shows the raw data of the packet, with the following fields highlighted:

- 0000 02 00 00 00 45 00 00 28 b5 f2 40 00 80 06 46 db ...E..(..@...F.
- 0010 7f 00 00 01 7f 00 00 01 2e e0 e4 64 fd fd 57 f2 ...d..W.
- 0020 b7 3f 08 2d 50 10 00 ff 88 31 00 00 ...?..P...1..

Annotations and explanations:

- Server 端監聽的 Port 58468
- Client 端隨機分配的 Port 12000
- 傳遞資料長度為 0
- Server 端的 Seq #，因為目前 Server 端目前沒有傳遞任何資料給 Client，表示下一個資料傳遞的序號編號為 1
- Ack = 4 表示 Server 已成功收到 Client 傳遞的 Seq 3 以前的所有訊息，並期待 Client 下一次發送從序號 4 開始的資料 (因為 seq = 1, len = 3, 所以 Ack = 1 + 3 = 4)
- Flag 中 ACK 的位元被標記，表示 Server 已收到 Client 傳遞的資料，且只有收到訊息，並沒有進行資料的處理

First Application data packet

Packet No.6: Server → Client

No.	Time	Source	Destination	Protocol	Length	Info
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.30763...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0

Frame 6: 46 bytes on wire (368 bits), 46 bytes captured (368 bits)

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

- 0100 = Version: 4 → 使用 IPv4 進行定址與路由
- 0101 = Header Length: 20
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 42
- Identification: 0xb5f3 (46579)
- 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6) → 使用 TCP 進行可靠傳輸
- Header Checksum: 0x46d8 [validation disabled]
- [Header checksum status: Unverified] → Server 端的 IP
- Source Address: 127.0.0.1
- Destination Address: 127.0.0.1 → Client 端的 IP
- [Stream index: 0]

Transmission Control Protocol, Src Port: 12000, Dst Port: 58468, S

Data (2 bytes)

0000 02 00 00 00 45 00 00 2a b5 f3 40 00 80 06 46 d8 ...E...*...@...F...
0010 7f 00 00 01 7f 00 00 01 2e e0 e4 64 fd fd 57 f2d...W...
0020 b7 3f 08 2d 50 18 00 ff 54 f2 00 00 33 35 ...?-P...T...35

First Application data packet

Packet No.6: Server → Client

The image shows a Wireshark packet capture of a TCP segment (Packet No. 6) sent from the server to the client. The packet list at the top shows the segment's details: No. 6, Time 4.30758..., Source 127.0.0.1, Destination 127.0.0.1, Protocol TCP, Length 46, and Info 12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2.

The packet details pane on the left shows the following fields:

- Transmission Control Protocol, Src
- Source Port: 12000
- Destination Port: 58468
- [Stream index: 0]
- [Stream Packet Number: 6]
- [Conversation completeness: Complete]
- [TCP Segment Len: 2]
- Sequence Number: 1 (relative sequence number)
- Sequence Number (raw): 4261238770
- [Next Sequence Number: 3 (relative sequence number)]
- Acknowledgment Number: 4 (relative ack number)
- Acknowledgment number (raw): 3074361389
- 0101 = Header Length: 20 bytes (5)
- Flags: 0x018 (PSH, ACK)
- Window: 255
- [Calculated window size: 65280]
- [Window size scaling factor: 256]
- Checksum: 0x54f2 [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- [Timestamps]
- [SEQ/ACK analysis]
- TCP payload (2 bytes)
- Data (2 bytes)

The packet bytes pane on the right shows the raw data in hexadecimal and ASCII. The ASCII column shows the data being transmitted: "E..*..@..F..d..W..?..P..T..35".

Annotations in Chinese explain the fields:

- Server 端監聽的 Port (Server side listening port) points to Source Port: 12000.
- Client 端隨機分配的 Port (Client side randomly assigned port) points to Destination Port: 58468.
- 傳遞資料長度為 2 (Data transfer length is 2) points to [TCP Segment Len: 2].
- Server 端的 Seq #, 表示目前傳遞給 Client 端的 Byte Stream 的起始序號 (Server side Seq #, indicating the starting sequence number of the Byte Stream currently being transferred to the Client side) points to Sequence Number: 1.
- Ack = 4 表示 Server 已成功收到 Client 傳遞的 Seq 3 以前的所有訊息, 並期待 Client 下一次發送從序號 4 開始的資料 (Ack = 4 indicates that the Server has successfully received all messages sent by the Client before Seq 3, and expects the Client to send data starting from Seq 4 next time) points to Acknowledgment Number: 4.
- Flag 中 ACK、PSH 的位元被標記, ACK 位元表示 Server 已收到先前 Client 端傳遞的資料, PSH 位元則是表示封包攜帶了應用層的資訊, 需要立即傳遞至應用程式進行資料處理 (In the flag, the ACK and PSH bits are marked. The ACK bit indicates that the Server has received the data previously sent by the Client, and the PSH bit indicates that the packet carries application layer information that needs to be immediately passed to the application for processing) points to Flags: 0x018 (PSH, ACK).

The packet bytes pane on the right shows the raw data in hexadecimal and ASCII. The ASCII column shows the data being transmitted: "E..*..@..F..d..W..?..P..T..35".

First Application data packet

Packet No.6: Server → Client

No.	Time	Source	Destination	Protocol	Length	Info
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.30763...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0

[Stream index: 0]
[Stream Packet Number: 6]
▸ [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 2]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 4261238770
[Next Sequence Number: 3 (relative sequence number)]
Acknowledgment Number: 4 (relative ack number)
Acknowledgment number (raw): 3074361389
0101 = Header Length: 20 bytes (5)
▸ Flags: 0x018 (PSH, ACK)
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0x54f2 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
▸ [Timestamps]
▸ [SEQ/ACK analysis]
TCP payload (2 bytes)
▾ Data (2 bytes)
Data: 3335
[Length: 2]

0000 02 00 00 00 45 00 00 2a b5 f3 40 00 80 06 46 d8 E . . * . . @ . . . F .
0010 7f 00 00 01 7f 00 00 01 2e e0 e4 64 fd fd 57 f2 d . W .
0020 b7 3f 08 2d 50 18 00 ff 54 f2 00 00 33 35 . ? . - P . . T . 35

Server 實際上傳遞給 Client 的資料

First Application data packet

Packet No.7: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.30763...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0

Frame 7: 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 40
- Identification: 0xb5f4 (46580)
- 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6)
- Header Checksum: 0x46d9 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 127.0.0.1
- Destination Address: 127.0.0.1
- [Stream index: 0]

Transmission Control Protocol, Src Port: 58468, Dst Port: 12000, Seq: 58468, Win: 65280, Len: 0

Hex dump: 0000 02 00 00 00 45 00 00 28 b5 f4 40 00 80 06 46 d9 ... E..(..@...F.
0010 7f 00 00 01 7f 00 00 01 e4 64 2e e0 b7 3f 08 2d d...?..
0020 fd fd 57 f4 50 10 00 ff 88 2f 00 00 ... W.P... ./..

Annotations:

- 使用 IPv4 進行定址與路由 (points to Version: 4)
- 使用 TCP 進行可靠傳輸 (points to Protocol: TCP (6))
- Client 端的 IP (points to Source Address: 127.0.0.1)
- Server 端的 IP (points to Destination Address: 127.0.0.1)

First Application data packet

Packet No.7: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
4	4.30734...	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.30741...	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.30758...	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.30763...	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 58468, Dst Port: 12000

[Stream index: 0]
[Stream Packet Number: 7]
[Conversation completeness: Complete]
[TCP Segment Len: 0]
Sequence Number: 4 (relative sequence number)
Sequence Number (raw): 3074361389
[Next Sequence Number: 4 (relative sequence number)]
Acknowledgment Number: 3 (relative ack number)
Acknowledgment number (raw): 4261238772
0101 = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0x882f [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]

Client 端隨機分配的 Port

Server 端監聽的 Port

傳遞資料長度為 0

Client 端的 Seq #，因為目前 Client 端目前沒有傳遞任何資料給 Server，表示下一個資料傳遞的序號編號為 4

Ack = 3 表示 Client 已成功收到 Server 傳遞的 Seq 2 以前的所有訊息，並期待 Server 下一次發送從序號 3 開始的資料（因為 seq = 1, len = 2，所以 Ack = 1 + 2 = 3）

Flag 中 ACK 的位元被標記，表示 Client 已收到 Server 傳遞的資料

Flags: 0x010 (ACK)

- 000. = Reserved: Not set
- ...0 = Accurate ECN: Not set
- ...0... .. = Congestion Window Reduced: Not set
- ...0... .. = ECN-Echo: Not set
- ...0... .. = Urgent: Not set
- ...1... .. = Acknowledgment: Set
- ...0... .. = Push: Not set
- ...0... .. = Reset: Not set
- ...0... .. = Syn: Not set
- ...0... .. = Fin: Not set

[TCP Flags:A....]

RTT (Round Trip Time)

- First Application data packet
(No.4~No.7)

$$\begin{aligned} \text{RTT} &= 4.307416700 - 4.307345100 \\ &= 0.000071600 \text{ s} \end{aligned}$$

4	4.307345100	127.0.0.1	127.0.0.1	TCP	47	58468 → 12000 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=3
5	4.307416700	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=1 Ack=4 Win=65280 Len=0
6	4.307586600	127.0.0.1	127.0.0.1	TCP	46	12000 → 58468 [PSH, ACK] Seq=1 Ack=4 Win=65280 Len=2
7	4.307632400	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=4 Ack=3 Win=65280 Len=0

Connection Close Packets

Packet No.16: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
16	16.481269400	127.0.0.1	127.0.0.1	TCP	48	58468 → 12000 [PSH, ACK] Seq=14 Ack=11 Win=65280 Len=4
17	16.481341200	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=11 Ack=18 Win=65280 Len=0
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0

Frame 16: 48 bytes on wire (384 bits), 48 bytes captured (384 bit)

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

- 0100 = Version: 4 → 使用 IPv4 進行定址與路由
- 0101 = Header Length: 20
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 44
- Identification: 0xb6cb (46795)
- 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6) → 使用 TCP 進行可靠傳輸
- Header Checksum: 0x45fe [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 127.0.0.1 → Client 端的 IP
- Destination Address: 127.0.0.1 → Server 端的 IP
- [Stream index: 0]

Transmission Control Protocol, Src Port: 58468, Dst Port: 12000,

Data (4 bytes)

0000 02 00 00 00 45 00 00 2c b6 cb 40 00 80 06 45 feE., ..@...E.
0010 7f 00 00 01 7f 00 00 01 e4 64 2e e0 b7 3f 08 37d...?·7
0020 fd fd 57 fc 50 18 00 ff b9 24 00 00 65 78 69 74 ...W·P...·\$.exit

Connection Close Packets

Packet No.16: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
16	16.481269400	127.0.0.1	127.0.0.1	TCP	48	58468 → 12000 [PSH, ACK] Seq=14 Ack=11 Win=65280 Len=4
17	16.481341200	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=11 Ack=18 Win=65280 Len=0
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0

Transmission Control Protocol, Seq=14, Len=4, Window=65280

Source Port: 58468 → Client 端隨機分配的 Port

Destination Port: 12000 → Server 端監聽的 Port

[Stream index: 0]

[Stream Packet Number: 16]

[Conversation completeness: 31]

[TCP Segment Len: 4] → 傳遞資料長度為 4

Sequence Number: 14 (relative sequence number) → Client 端的 Seq #, 表示目前傳遞給 Server 端的 Byte Stream 的起始序號

Sequence Number (raw): 3074361399

[Next Sequence Number: 18 (relative sequence number)]

Acknowledgment Number: 11 (relative ack number) → Ack = 11 表示 Client 已成功收到 Server 傳遞的 Seq 10 以前的所有訊息, 並期待 Server 下一次發送從序號 11 開始的資料

Acknowledgment number (raw): 4261238780

0101 = Header Length: 20 bytes (5)

Flags: 0x018 (PSH, ACK) → Flag 中 ACK、PSH 的位元被標記, ACK 位元表示 Client 已收到先前 Server 端傳遞的資料, PSH 位元則是表示封包攜帶了應用層的資訊, 需要立即傳遞至應用程式進行資料處理

Window: 255

[Calculated window size: 65280]

[Window size scaling factor: 256]

Checksum: 0xb924 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

[SEQ/ACK analysis]

TCP payload (4 bytes)

Data (4 bytes)

0000 02 00 00 00 45 00 00 2c b6 cb 40 00 80 06 45 fe ...E., .@...E.
0010 7f 00 00 01 7f 00 00 01 e4 64 2e e0 b7 3f 08 37d...?.7
0020 fd fd 57 fc 50 18 00 ff b9 24 00 00 65 78 69 74 ...W.P...\$.exit

Flags: 0x018 (PSH, ACK)

000. = Reserved: Not set

...0 = Accurate ECN: Not set

...0... = Congestion Window Reduced: Not set

...0... = ECN-Echo: Not set

...0... = Urgent: Not set

...1... = Acknowledgment: Set

...1... = Push: Set

...0... = Reset: Not set

...0... = Syn: Not set

...0... = Fin: Not set

[TCP Flags:AP...]

First Application data packet

Packet No.16: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
16	16.481269400	127.0.0.1	127.0.0.1	TCP	48	58468 → 12000 [PSH, ACK] Seq=14 Ack=11 Win=65280 Len=4
17	16.481341200	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=11 Ack=18 Win=65280 Len=0
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0
20	16.481711100	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [FIN, ACK] Seq=18 Ack=12 Win=65280 Len=0

000. = Reserved: Not set

...0 = Accurate ECN: Not set

.... 0... = Congestion Window Reduced: Not set

.... .0.. = ECN-Echo: Not set

.... ..0. = Urgent: Not set

.... ...1 = Acknowledgment: Set

.... 1... = Push: Set

....0.. = Reset: Not set

....0. = Syn: Not set

....0 = Fin: Not set

[TCP Flags:AP...]

Window: 255

[Calculated window size: 65280]

[Window size scaling factor: 256]

Checksum: 0xb924 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

▸ [Timestamps]

▸ [SEQ/ACK analysis]

TCP payload (4 bytes)

▾ Data (4 bytes)

Data: 65786974

[Length: 4]

0000 02 00 00 00 45 00 00 2c b6 cb 40 00 80 06 45 feE., ..@...E.

0010 7f 00 00 01 7f 00 00 01 e4 64 2e e0 b7 3f 08 37d...?7

0020 fd fd 57 fc 50 18 00 ff b9 24 00 00 65 78 69 74 ..W.P...\$.exit

Client 實際上傳遞給 Server 的資料

Connection Close Packets

Packet No.17: Server → Client

No.	Time	Source	Destination	Protocol	Length	Info
✓ 16	16.481269400	127.0.0.1	127.0.0.1	TCP	48	58468 → 12000 [PSH, ACK] Seq=14 Ack=11 Win=65280 Len=4
17	16.481341200	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=11 Ack=18 Win=65280 Len=0
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0

▶ Frame 17: 44 bytes on wire (352 bits), 44 bytes captured (352 bit)

▶ Null/Loopback

▼ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 40

Identification: 0xb6cc (46796)

▶ 010. = Flags: 0x2, Don't fragment

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 128

Protocol: TCP (6)

Header Checksum: 0x4601 [validation disabled]

[Header checksum status: Unverified]

Source Address: 127.0.0.1

Destination Address: 127.0.0.1

[Stream index: 0]

▶ Transmission Control Protocol, Src Port: 12000, Dst Port: 58468,

0000 02 00 00 00 45 00 00 28 b6 cc 40 00 80 06 46 01 ...E..(..@...F.

0010 7f 00 00 01 7f 00 00 01 2e e0 e4 64 fd fd 57 fcd..W.

0020 b7 3f 08 3b 50 10 00 ff 88 19 00 00 ...?.;P... ..

使用 IPv4 進行定址與路由

使用 TCP 進行可靠傳輸

Server 端的 IP

Client 端的 IP

Connection Close Packets

Packet No.17: Server → Client

No.	Time	Source	Destination	Protocol	Length	Info
✓ 16	16.481269400	127.0.0.1	127.0.0.1	TCP	48	58468 → 12000 [PSH, ACK] Seq=14 Ack=11 Win=65280 Len=4
17	16.481341200	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=11 Ack=18 Win=65280 Len=0
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 12000, Dst Port: 58468

Source Port: 12000 → Server 端監聽的 Port

Destination Port: 58468 → Client 端隨機分配的 Port

[Stream index: 0]

[Stream Packet Number: 17]

[Conversation completeness: 0 (31)]

[TCP Segment Len: 0] → 傳遞資料長度為 0

Sequence Number: 11 (relative sequence number) → Server 端的 Seq #，因為目前 Server 端目前沒有傳遞任何資料給 Client，表示下一個資料傳遞的序號編號為 11

Sequence Number (raw): 4261238780

[Next Sequence Number: 11 (relative sequence number)]

Acknowledgment Number: 18 (relative ack number) → Ack = 18 表示 Server 已成功收到 Client 傳遞的 Seq 17 以前的所有訊息，並期待 Server 下一次發送從序號 18 開始的資料

Acknowledgment number (raw): 3074361403

0101 = Header Length: 20 bytes (5)

Flags: 0x010 (ACK) → Flag 中 ACK，ACK 位元表示 Server 已收到先前 Client 端傳遞想要終止連結的訊息

Window: 255

[Calculated window size: 65280]

[Window size scaling factor: 256]

Checksum: 0x8819 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

[SEQ/ACK analysis]

Flags: 0x010 (ACK)

000. = Reserved: Not set

...0 = Accurate ECN: Not set

... 0... = Congestion Window Reduced: Not set

... .0.. = ECN-Echo: Not set

... 0... = Urgent: Not set

... ..1... = Acknowledgment: Set

... ..0... = Push: Not set

... ..0.. = Reset: Not set

... ..0.. = Syn: Not set

... ..0.. = Fin: Not set

[TCP Flags:A.....]

Connection Close Packets

Packet No.18: Server → Client

No.	Time	Source	Destination	Protocol	Length	Info
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0
20	16.481711100	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [FIN, ACK] Seq=18 Ack=12 Win=65280 Len=0
21	16.481767000	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=12 Ack=19 Win=65280 Len=0

▶ Frame 18: 44 bytes on wire (352 bits), 44 bytes captured (352 bits)	0000	02 00 00 00 45 00 00 28	b6 cd 40 00 80 06 46 00	E..(..@...F.
▶ Null/Loopback	0010	7f 00 00 01 7f 00 00 01	2e e0 e4 64 fd fd 57 fc	d.W.
▼ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1	0020	b7 3f 08 3b 50 11 00 ff	88 18 00 00	.?.;P... ..
0100 = Version: 4				
.... 0101 = Header Length: 20				
▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)				
Total Length: 40				
Identification: 0xb6cd (46797)				
▶ 010. = Flags: 0x2, Don't fragment				
...0 0000 0000 0000 = Fragment Offset: 0				
Time to Live: 128				
<u>Protocol: TCP (6)</u>				
Header Checksum: 0x4600 [validation disabled]				
[Header checksum status: Unverified]				
<u>Source Address: 127.0.0.1</u>				
<u>Destination Address: 127.0.0.1</u>				
[Stream index: 0]				
▶ Transmission Control Protocol, Src Port: 12000, Dst Port: 58468, S				

Connection Close Packets

Packet No.18: Server → Client

No.	Time	Source	Destination	Protocol	Length	Info
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0
20	16.481711100	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [FIN, ACK] Seq=18 Ack=12 Win=65280 Len=0
21	16.481767000	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=12 Ack=19 Win=65280 Len=0

Frame 18: 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 12000, Dst Port: 58468, Seq: 11, Len: 0

Source Port: 12000 → Server 端監聽的 Port
Destination Port: 58468 → Client 端隨機分配的 Port
[Stream index: 0]
[Stream Packet Number: 18]
[Conversation completeness: 0] → 傳遞資料長度為 0
[TCP Segment Len: 0] → 傳遞資料長度為 0
Sequence Number: 11 (relative sequence number) → Server 端的 Seq #, 表示目前傳遞給 Client 端的 Byte Stream 的起始序號
Sequence Number (raw): 4261238780
[Next Sequence Number: 12 (relative sequence number)]
Acknowledgment Number: 18 (relative ack number) → Ack = 18 表示 Server 已成功收到 Client 傳遞的 Seq 17 以前的所有訊息, 並期待 Client 下一次發送從序號 18 開始的資料
Acknowledgment number (raw): 3074361403
0101 = Header Length: 20 bytes (5)
Flags: 0x011 (FIN, ACK) → Flag 中 FIN、ACK 的位元被標記, ACK 位元表示 Server 已收到先前 Client 端傳遞的資料, FIN 位元則是表示 Server 端已經完成資料的傳送, 準備進行連線的關閉
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0x8818 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]

0000 02 00 00 00 45 00 00 28 b6 cd 40 00 80 06 46 00 ...E..(..@...F.
0010 7f 00 00 01 7f 00 00 01 2e e0 e4 64 fd fd 57 fcd..W.
0020 b7 3f 08 3b 50 11 00 ff 88 18 00 00?.;P... ..

Flags: 0x011 (FIN, ACK)
000. = Reserved: Not set
...0 = Accurate ECN: Not set
.... 0... = Congestion Window Reduced: Not set
.... 0... = ECN-Echo: Not set
.... 0... = Urgent: Not set
.... 1... = Acknowledgment: Set
.... 0... = Push: Not set
.... 0... = Reset: Not set
.... 0... = Syn: Not set
.... 1... = Fin: Set
[TCP Flags: 0x011 (FIN, ACK)]

Connection Close Packets

Packet No.19: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
✓ 18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0
20	16.481711100	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [FIN, ACK] Seq=18 Ack=12 Win=65280 Len=0
21	16.481767000	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=12 Ack=19 Win=65280 Len=0

Frame 19: 44 bytes on wire (352 bits), 44 bytes captured (352 bit)

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

- 0100 = Version: 4 → 使用 IPv4 進行定址與路由
- 0101 = Header Length: 20
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 40
- Identification: 0xb6ce (46798)
- 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6) → 使用 TCP 進行可靠傳輸
- Header Checksum: 0x45ff [validation disabled]
- [Header checksum status: Unverified] → Client 端的 IP
- Source Address: 127.0.0.1
- Destination Address: 127.0.0.1 → Server 端的 IP
- [Stream index: 0]

Transmission Control Protocol, Src Port: 58468, Dst Port: 12000,

0000 02 00 00 00 45 00 00 28 b6 ce 40 00 80 06 45 ffE..(..@...E.
0010 7f 00 00 01 7f 00 00 01 e4 64 2e e0 b7 3f 08 3bd...?.;
0020 fd fd 57 fd 50 10 00 ff 88 18 00 00 ..W.P... ..

Connection Close Packets

Packet No.19: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0
20	16.481711100	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [FIN, ACK] Seq=18 Ack=12 Win=65280 Len=0
21	16.481767000	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=12 Ack=19 Win=65280 Len=0

Frame 19: 44 bytes on wire (352 bits), 44 bytes captured (352 b)

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 58468, Destination Port: 12000

[Stream index: 0]

[Stream Packet Number: 19]

[Conversation completeness: 0]

[TCP Segment Len: 0]

Sequence Number: 18 (relative sequence number)

Sequence Number (raw): 3074361403

[Next Sequence Number: 18 (relative sequence number)]

Acknowledgment Number: 12 (relative ack number)

Acknowledgment number (raw): 4261238781

0101 = Header Length: 20 bytes (5)

Flags: 0x010 (ACK)

Window: 255

[Calculated window size: 65280]

[Window size scaling factor: 256]

Checksum: 0x8818 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

Client 端隨機分配的 Port

Server 端監聽的 Port

傳遞資料長度為 0

Client 端的 Seq #，因為目前 Client 端目前沒有傳遞任何資料給 Server，表示下一個資料傳遞的序號編號為 12

Ack = 12 表示 Client 已成功收到 Server 傳遞的 Seq 11 以前的所有訊息，並期待 Server 下一次發送從序號 12 開始的資料

Flag 中 ACK 的位元被標記，ACK 位元表示 Client 已確認收到 Server 傳送的關閉連線之訊息

Flags: 0x010 (ACK)

000. = Reserved: Not set

...0. = Accurate ECN: Not set

...0. = Congestion Window Reduced: Not set

...0. = ECN-Echo: Not set

...0. = Urgent: Not set

...1. = Acknowledgment: Set

...0. = Push: Not set

...0. = Reset: Not set

...0. = Syn: Not set

...0. = Fin: Not set

[TCP Flags:A....]

Connection Close Packets

Packet No.20: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0
20	16.481711100	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [FIN, ACK] Seq=18 Ack=12 Win=65280 Len=0
21	16.481767000	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=12 Ack=19 Win=65280 Len=0

Frame 20: 44 bytes on wire (352 bits), 44 bytes captured (352 bit)

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

0100 = Version: 4

.... 0101 = Header Length: 20

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 40

Identification: 0xb6cf (46799)

010. = Flags: 0x2, Don't fragment

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 128

Protocol: TCP (6)

Header Checksum: 0x45fe [validation disabled]

[Header checksum status: Unverified]

Source Address: 127.0.0.1

Destination Address: 127.0.0.1

[Stream index: 0]

Transmission Control Protocol, Src Port: 58468, Dst Port: 12000,

0000 02 00 00 00 45 00 00 28 b6 cf 40 00 80 06 45 fe

0010 7f 00 00 01 7f 00 00 01 e4 64 2e e0 b7 3f 08 3b

0020 fd fd 57 fd 50 11 00 ff 88 17 00 00

.....E..(..@...E.

.....d...?..;

..W.P... ..

使用 IPv4 進行定址與路由

使用 TCP 進行可靠傳輸

Client 端的 IP

Server 端的 IP

Connection Close Packets

Packet No.20: Client → Server

No.	Time	Source	Destination	Protocol	Length	Info
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0
20	16.481711100	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [FIN, ACK] Seq=18 Ack=12 Win=65280 Len=0
21	16.481767000	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=12 Ack=19 Win=65280 Len=0

Frame 20: 44 bytes on wire (352 bits), 44 bytes captured (352 bit) on interface 0

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Seq: 58468, Len: 0

Source Port: 58468 (Client 端隨機分配的 Port)

Destination Port: 12000 (Server 端監聽的 Port)

[Stream index: 0]

[Stream Packet Number: 20]

[Conversation completeness: 0]

[TCP Segment Len: 0] (傳遞資料長度為 0)

Sequence Number: 18 (relative sequence number)

Sequence Number (raw): 3074361403

[Next Sequence Number: 19 (relative sequence number)]

Acknowledgment Number: 12 (relative ack number)

Acknowledgment number (raw): 4261238781

0101 = Header Length: 20 bytes (5)

Flags: 0x011 (FIN, ACK)

Window: 255

[Calculated window size: 65280]

[Window size scaling factor: 256]

Checksum: 0x8817 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

Flags: 0x011 (FIN, ACK)

000. = Reserved: Not set

...0 = Accurate ECN: Not set

...0 = Congestion Window Reduced: Not set

...0 = ECN-Echo: Not set

...0 = Urgent: Not set

...1 = Acknowledgment: Set

...0 = Push: Not set

...0 = Reset: Not set

...0 = Syn: Not set

...1 = Fin: Set

Connection Close Packets

Packet No.21: Server → Client

No.	Time	Source	Destination	Protocol	Length	Info
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0
20	16.481711100	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [FIN, ACK] Seq=18 Ack=12 Win=65280 Len=0
21	16.481767000	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=12 Ack=19 Win=65280 Len=0

▶ Frame 21: 44 bytes on wire (352 bits), 44 bytes captured (352 bit	0000	02 00 00 00 45 00 00 28 b6 d0 40 00 80 06 45 fd	...E..(..@...E.
▶ Null/Loopback	0010	7f 00 00 01 7f 00 00 01 2e e0 e4 64 fd fd 57 fd	d..W.
▼ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1	0020	b7 3f 08 3c 50 10 00 ff 88 17 00 00	..?.<P... ..
0100 = Version: 4			
.... 0101 = Header Length: 20			
▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)			
Total Length: 40			
Identification: 0xb6d0 (46800)			
▶ 010. = Flags: 0x2, Don't fragment			
...0 0000 0000 0000 = Fragment Offset: 0			
Time to Live: 128			
Protocol: TCP (6)			
Header Checksum: 0x45fd [validation disabled]			
[Header checksum status: Unverified]			
Source Address: 127.0.0.1			
Destination Address: 127.0.0.1			
[Stream index: 0]			
▶ Transmission Control Protocol, Src Port: 12000, Dst Port: 58468,			

Connection Close Packets

Packet No.21: Server → Client

No.	Time	Source	Destination	Protocol	Length	Info
18	16.481484600	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [FIN, ACK] Seq=11 Ack=18 Win=65280 Len=0
19	16.481528200	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [ACK] Seq=18 Ack=12 Win=65280 Len=0
20	16.481711100	127.0.0.1	127.0.0.1	TCP	44	58468 → 12000 [FIN, ACK] Seq=18 Ack=12 Win=65280 Len=0
21	16.481767000	127.0.0.1	127.0.0.1	TCP	44	12000 → 58468 [ACK] Seq=12 Ack=19 Win=65280 Len=0

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Seq: 12000, Len: 0

Source Port: 12000 → Server 端監聽的 Port

Destination Port: 58468 → Client 端隨機分配的 Port

[Stream index: 0]

[Stream Packet Number: 21]

[Conversation completeness: 1]

[TCP Segment Len: 0] → 傳遞資料長度為 0

Sequence Number: 12 (relative sequence number) → Server 端的 Seq #，因為目前 Server 端目前沒有傳遞任何資料給 Client，表示下一個資料傳遞的序號編號為 12

Sequence Number (raw): 4261238781

[Next Sequence Number: 12 (relative sequence number)]

Acknowledgment Number: 19 (relative ack number) → Ack = 19 表示 Server 已成功收到 Client 傳遞的 Seq 18 以前的所有訊息，並期待 Client 下一次發送從序號 19 開始的資料

Acknowledgment number (raw): 3074361404

0101 = Header Length: 20 bytes (5)

Flags: 0x010 (ACK) → Flag 中 ACK 的位元被標記，ACK 位元表示 Server 已確認收到 Client 傳送的關閉連線之訊息

Window: 255

[Calculated window size: 65280]

[Window size scaling factor: 256]

Checksum: 0x8817 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

[SEQ/ACK analysis]

0000 02 00 00 00 45 00 00 28 b6 d0 40 00 80 06 45 fd ...E..(..@..E..

0010 7f 00 00 01 7f 00 00 01 2e e0 e4 64 fd fd 57 fdd..W..

0020 b7 3f 08 3c 50 10 00 ff 88 17 00 00 ...?<P.....

Flags: 0x010 (ACK)

000. = Reserved: Not set

...0 = Accurate ECN: Not set

....0... = Congestion Window Reduced: Not set

....0... = ECN-Echo: Not set

....0... = Urgent: Not set

....0... = Acknowledgment: Set

....0... = Push: Not set

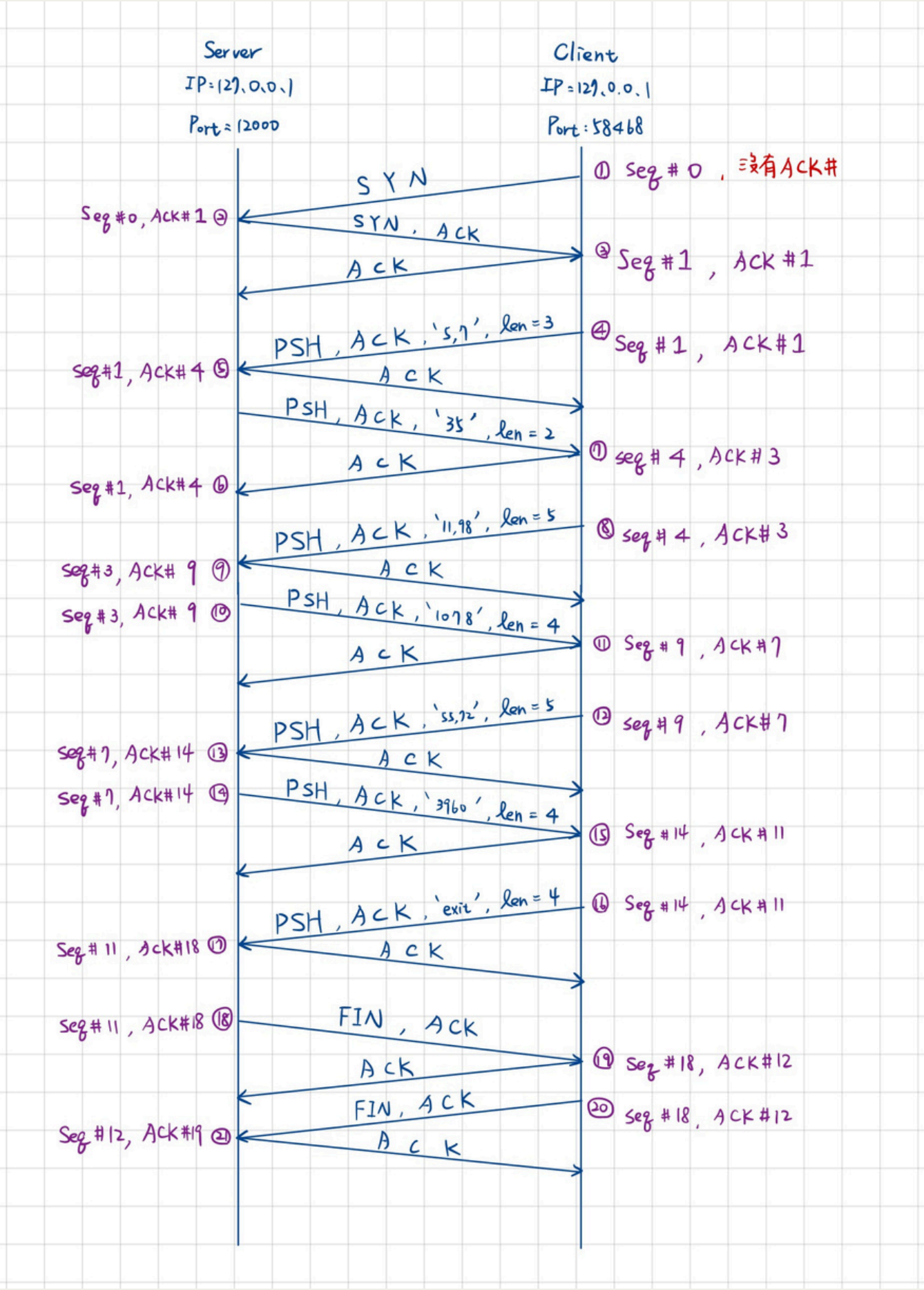
....0... = Reset: Not set

....0... = Syn: Not set

....0... = Fin: Not set

[TCP Flags:A.....]

Time Sequence Diagram



Thank you!
